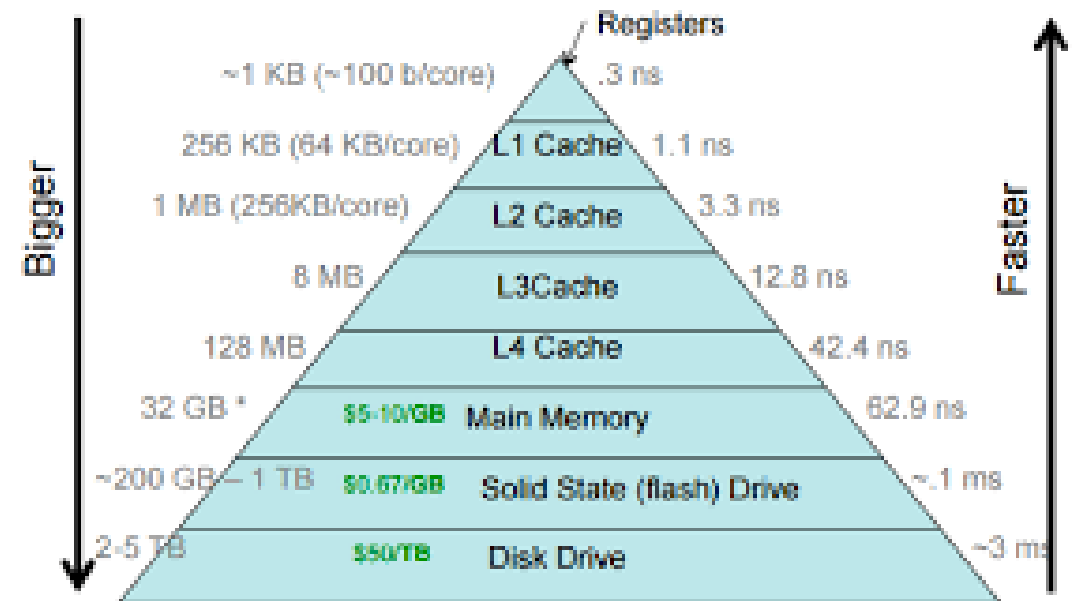


Learning memory patterns

Mar 10 2026

Memory hierarchy 101

- Computation faster than memory access
- Cache frequently used data
- Types of cache misses:
 - Compulsory
 - Conflict
 - Capacity



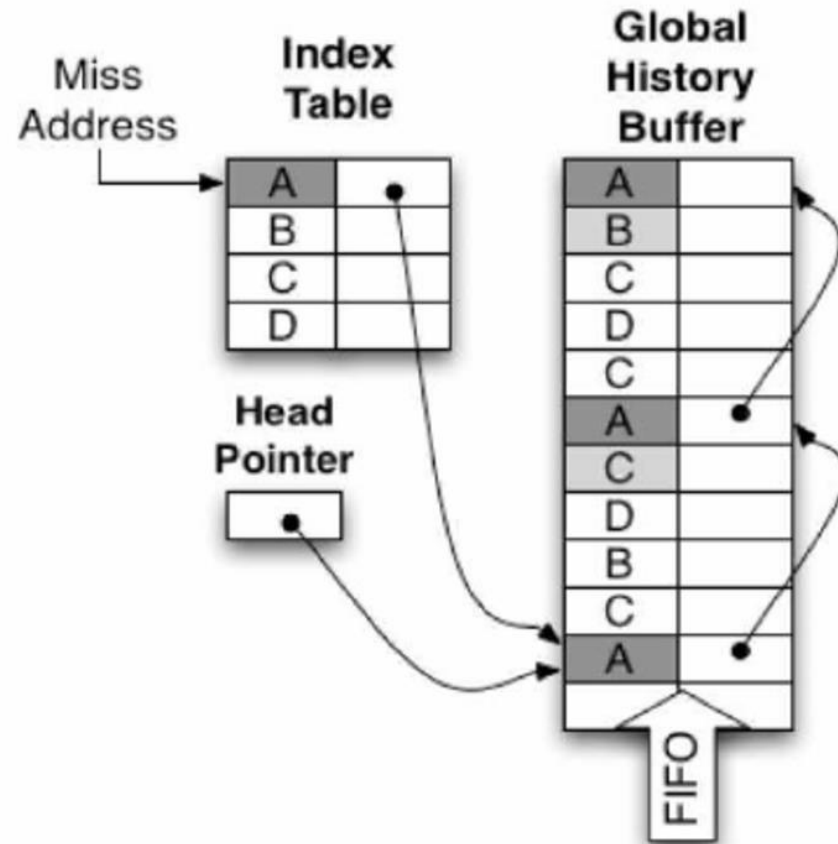
Prefetching

- Three questions:
 - What to prefetch?
 - When to prefetch?
 - How much to prefetch?
- Accuracy
Timeliness
Coverage
- Balance accuracy with cache pollution

Types of prefetching

- Software
 - Prefetch hints
 - Helper threading
- Hardware
 - Next-line
 - Stride
 - Markov/correlation
 - Delta-based
 - Spatial
 - Pointer-based

GHB prefetcher



LSTM

